SCIP PANELS

The Future of Smart, Resilient & Profitable Construction

Presented by: **David Coons** | Coons Construction Inc

Southern Colorado Real Estate Investors Group | 1-Hour Educational Presentation

SCIP Panels Supplied by: Total Integrated Panel Systems (TIPs) · Morgan, Utah

Your Next 60 Minutes

Today we'll look at a construction system that combines structural performance, energy efficiency, and long-term value.

What Is a SCIP Panel?

Structural Concrete Insulated Panel — four words that every builder, investor and developer should know

Imagine a wall system that combines structure, insulation, and durability into a single assembly.

A Brief (But Fascinating) History

- Early 1960s: Concept born in Europe — the idea was right, machines didn't exist yet
- 1967: Victor Weismann patents the first SCIP system in Pasadena, CA (US3305991)
- 1970s–80s: Austrian company EVG invents high-speed automated welding machinery
- 1990s–2000s: Adopted across Europe, Middle East, and South America at scale
- Today: TIPs (Morgan, Utah) brings the most advanced SCIP system to the US market

Decoding the Acronym

S — **STRUCTURAL**
Carries load — walls, floors, roofs. No extra framing needed.

C — **CONCRETE**
Shotcrete applied on both sides = monolithic shell strength.

I — **INSULATED**
4" EPS foam core = continuous thermal + acoustic insulation.

P — **PANEL**
Pre-engineered, lightweight, fast to ship and install anywhere.

How SCIP Panels Are Made

Every component serves a purpose – and together they create a single structural system.

Welded Wire Mesh (WWR)

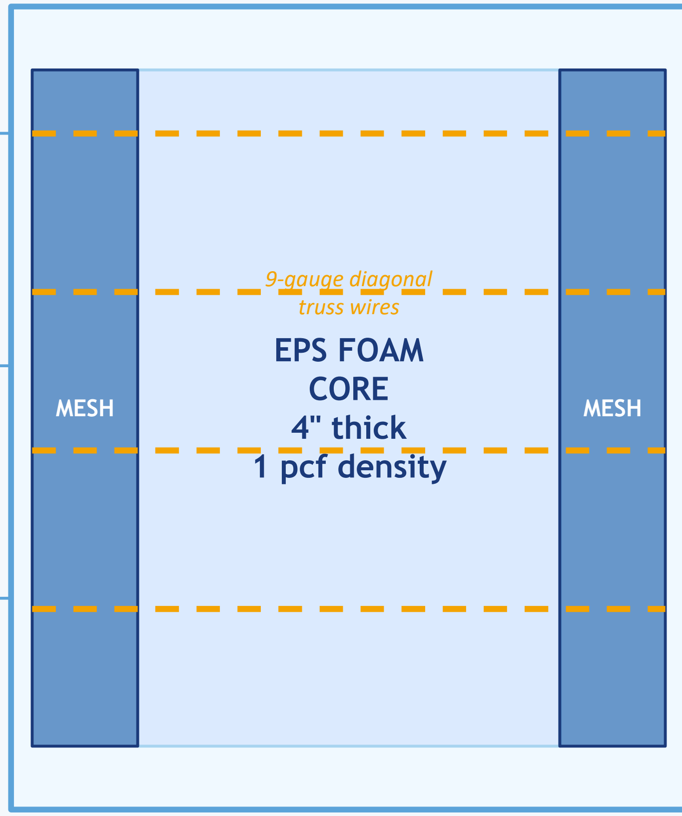
11-gauge galvanized steel
2"×2" grid, ASTM A1064
Fy = 60 ksi yield strength

EPS Foam Core

4" expanded polystyrene
1 pcf density, ASTM C578
Type I — zero off-gassing

Diagonal Truss Wires

9-gauge galvanized steel
Pyramidal pattern (TIPs)
Fy = 65 ksi — creates 3D composite



Shotcrete Shells

Min 3,500 psi (f'c)
¾" over mesh on each side
Applied on-site — bonds monolithically

Finished Wall System

Replaces: framing + insulation
+ drywall + exterior sheathing
50% lighter than solid concrete

TIPs Proprietary Advantage

Proprietary pyramidal truss geometry designed for superior composite performance

Why We Chose TIPs

Total Integrated Panel Systems · Morgan, Utah · Engineered for High Performance

50%

Less Concrete
vs. traditional tilt-up

85%

Less Rebar
without sacrificing strength

100%

Fully Composite Structural System

#1

One of the strongest SCIP systems available in North America.

🏗 Factory-Engineered

EVG Austrian automated welding machinery. Zero-waste production in Morgan, Utah.

🔺 Pyramidal Truss Design

TIPs' proprietary geometry creates the strongest SCIP composite available.

📐 Pre-Fabricated Openings

Windows and doors pre-cut at factory — no on-site guesswork.

🌡 4" EPS Fully Integrated

Insulation pre-connected, not added later — saves time and ensures performance.

🏠 Multi-Application

Residential, commercial, pools, sound walls, retaining walls — one system.

🤝 TCA Member

Tilt-Up Concrete Association certified — serves all 50 US states and Canada.

From Raw Materials to Ready-to-Build

How TIPs panels go from wire spools to your jobsite

STEP 01

EPS Foam Cutting

Type I expanded polystyrene blocks are precision-cut to exact thickness (typically 4"). Density: 1 ±0 pcf. Zero toxic off-gassing. ASTM C578 compliant.

STEP 02

Wire Mesh Fabrication

Galvanized 11-gauge wire welded into 2×2" grid sheets. Both faces identical. ASTM A1064, 60 ksi yield. Corrosion-resistant galvanized coating.

STEP 03

Automated EVG Welding

Austrian EVG high-speed welding line spaces and punches 9-gauge diagonal truss wires through the EPS and welds them to both mesh faces simultaneously. Zero waste.

STEP 04

TIPs Pyramidal Assembly

TIPs' proprietary pyramidal truss geometry is applied — creating the strongest 3D composite SCIP structure on the market. This is their competitive advantage.

STEP 05

Pre-Fabrication

Openings for windows and doors cut per construction plans. Panels cut to job-specific height. Inspected and bundled for delivery.

STEP 06

On-Site Shotcrete

Contractor applies 3,500 psi shotcrete (¾" over mesh each side). Utilities roughed in beforehand. Result: monolithic composite structural wall.

SCIP is a Global Industry

SCIP isn't new. It's been used around the world for decades.

Global Suppliers & Manufacturers

TIPs (Our Partner)

Morgan, Utah, USA

Pyramidal composite — strongest SCIP. TCA member. Ships nationwide.

Innstruct / ISS

Morgan, Utah, USA

Uses EVG machinery. Supplies residential & commercial across western US.

Fortified Structural Solutions (FSS)

United States

ICC-ES ESR-5623 listed. Uses Schnell Home machinery. Residential focus.

Carmelo / GCT

Puerto Rico / US

Hurricane-tested to 283 mph winds. IBC Section 722 fire rating. Multi-family.

Concrewall

International

European heritage system. Used in residential and commercial projects globally.

RSG-3D

International

3D panel manufacturer serving construction in multiple regions worldwide.

Tridipanel

International

Popular in Hawaii eco-construction; EPS with recycled content and 3D truss.

EVG (Machinery MFR)

Austria

Invented the automated production machinery that made modern SCIP possible.

VERO / Others

Europe / Middle East

Widely used in seismic zones of southern Europe, Turkey, and Middle East.

Performance Data That Matters

Fact-checked statistics from manufacturers, engineers, and independent testing

R-16+
Mass Wall Thermal Performance
(4" EPS continuous insulation)

283 mph
Wind Resistance Tested
(Carmelo GCT TER lab results)

4-Hour
Max Fire Rating
IBC Sec. 722 (with added mortar)

.4g
Seismic Resistance
(~7.5 Richter scale)

50-80%
Heating/Cooling Savings
vs. traditional construction

Performance Comparison: SCIP vs. The Competition

Metric	Wood Frame	ICF	Trad. Tilt-Up	SCIP (TIPs)
R-Value (Wall)	R-13 to R-21	R-22 to R-26	R-0 (no insulation)	R-16+ continuous
Fire Rating	~1 hour	2–4 hours	Variable	1–4 hours (IBC 722)
Wind Resistance	~130 mph	200+ mph	200+ mph	200–283 mph tested
Seismic Resistance	Moderate	High	High	High (.4g / ~7.5R)
Wall Weight	~15 psf	~60–80 psf	~75 psf	~35 psf (50% lighter)
Build Speed	Slow	Moderate	Moderate	Fast (pre-fab)

Sources & References

Every claim in this presentation is backed by published research, manufacturer data, and independent testing

Panel Composition & Engineering

- Mashal, M., Heath, M.D., Farrell, J. (2022). "The Structural Concrete Insulated Panel (SCIP)." PCI Journal, Mar–Apr 2022. DOI via pci.org
- Mashal, M. et al. (2019). "SCIPs: An Alternative Construction Technology in Seismic Regions." ResearchGate / ASCE. DOI:10.1061
- ICC-ES ESR-5623 — Fortified Structural Solutions SCIP Panel Evaluation Report. icc-es.org
- TIPs (Total Integrated Panel Systems) Technical Specification Guide. tipspanel.com · Morgan, UT 84050
- Innstruct / ISS SCIP Introduction Manual (2021). 375 East 400 North, Morgan, Utah 84050

Fire, Wind & Seismic Performance

- Carmelo / GCT SCIP Technical Evaluation Report (TER) — fire rating IBC Sec. 722, wind cycles to 283 mph. carmelo.com
- IBC Section 722 — Determination of Fire Resistance Ratings. International Building Code (current edition)
- ASCE 7 Section 12.2.1 — Seismic Design Requirements for Concrete Structures
- ASTM C578 — Standard Specification for Rigid Cellular Polystyrene Thermal Insulation (EPS core compliance)
- ASTM A1064 — Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement (WWR compliance)

Energy Performance & R-Value

- Thermocore SIP documented case study: 3,573 sq ft home near Green Bay, WI — $3,265/yr savings. thermocore.com
- SIPA (Structural Insulated Panel Association) — SIP True Cost Bidding Tool (2023). sips.org
- U.S. Dept. of Energy / Sam Rashkin — Housing 2.0: A Disruption Survival Guide. Referenced in SIPA STCBT
- ACME Panel — "Energy Efficiency & R Values": 33% smaller HVAC, $30,000 savings over mortgage period. acmepanel.com
- Hefty Homes (Mark Anderson Design/Build) — SCIP system 50–80% heating cost reduction. markandersondesignbuild.com

Construction Cost Data — Colorado

- EVstudio — "Home Building Costs in the Rocky Mountain Region" (2024). $150–$800/sqft CO range. evstudio.com
- Sidney Aulds Building Studio — "2025 Colorado Construction Costs." $200–$500/sqft statewide avg. sidneyauldsbuildingstudio.com
- University of Colorado Boulder Honors Thesis — "The True Cost of SIPs" (CU Scholar ID: pv63g086f). scholar.colorado.edu
- LawnStarter — "How Much Do SIP Panels Cost in 2026?" $7–$12/sqft panel material range. lawnstarter.com

Manufacturer & Industry Sources

- TIPs (Total Integrated Panel Systems) — tipspanel.com · TCA Member Profile: tilt-up.org
- Utah SBDC — Total Integrated Panel Systems Success Story. utahsbdc.org
- Focus Utah & Idaho — "Innovative Structural Solutions Plant Tour" (Oct 2020). focusutah.com
- Fortified Structural Solutions — FSS SCIP Product Catalog & ICC-ES ESR-5623. fortifiedstructuralsolutions.com
- Carmelo / GCT — SCIP panel specs, bullet resistance NIJ IIIA, hurricane/seismic TER. carmelo.com
- HOUSEI — SCIP system overview, 200+ mph wind test documentation. housei.io/scip

Cost Comparison: The Real Numbers

Because cost is usually the first question people ask.

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The question isn't whether SCIP costs more. The question is what you're comparing it to.

Colorado New Construction Cost per Square Foot — 2025 Estimates

Sources: EVstudio.com · Sidney Aulds Building Studio · TIPs tipspanel.com · ISS Innstruct Manual (2021) · GreenBuildingTalk.com

Construction Method	Low Estimate ($/sqft)	High Estimate ($/sqft)	Typical Range	Insulation Included?	Notes
Wood Frame (Production)	$150	$180	$150–$180	No — add $15–25/sqft	Source: EVstudio 2024; Denver/CS market
Wood Frame (Custom)	$250	$400	$250–$400	No — add $15–25/sqft	Source: EVstudio 2024; custom/luxury
ICF Construction	$230	$380	$230–$380	Yes — integral	~10–20% premium over wood; higher labor cost
Conventional Tilt-Up	$180	$280	$180–$280	No — add separately	Source: TCA benchmarks; no insulation in panel
SCIP with TIPs Panels ★	**$175**	**$300**	**$175–$300**	**Yes — 4" EPS integral**	**Structure + insulation in one; –85% rebar, –50% concrete vs. tilt-up**

⚡ **Important:** The table reflects upfront construction cost only. SCIP's documented 50–80% energy savings, 85% less rebar vs. tilt-up, and 50% less concrete — plus a 100-year lifespan — make total cost of ownership significantly lower.

The Investor Case for SCIP

Why every investor in this room should be asking about SCIP on their next project

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The question isn't what SCIP costs to build. The question is what the building costs to own.

Lower Operating Costs = Higher NOI

- 50–80% reduction in HVAC energy = more net income
- Documented: $3,265/yr savings on 3,500 sq ft
- Smaller HVAC systems = lower CapEx at build
- Tenants LOVE lower utility bills → lower vacancy

Insurance Advantages

- Fire-rated up to 4 hours → potential premium reduction
- Wind-tested to 200–283 mph → major in hail/wind zones
- Seismic-rated → Colorado market relevant
- Bullet-resistant (Federal building standard NIJ Level IIIA)

Build Speed = Cost Savings

- Pre-fabricated panels = faster framing phase
- Openings pre-cut at factory = less on-site labor
- Shorter construction loan period = less interest
- One system replaces framing + insulation + exterior sheathing

Resale & Appraisal Value

- Energy-efficient buildings command 4–6% premium
- LEED certification pathway — assists with grants/tax credits
- Meeting VA, FHA, HUD thermal requirements
- Concrete/steel structure = 100+ year building lifespan

What Can You Build with SCIP?

The short answer: almost everything. The full answer: even better.

🏠 Single-Family Homes

Complete structural wall system replaces framing + insulation. Faster build, lower energy bills, higher durability. VA/FHA/HUD thermal compliant.

🏘️ Multi-Family / ADUs / Townhomes

Excellent for ADU projects (bonus: lower noise transmission between units). Stack up to 4 stories with standard SCIP without added structural systems.

🏢 Commercial Buildings

Used for warehouses, offices, retail. Tilt-up capability with TIPS means large commercial projects benefit from same concrete tilt-up aesthetics at lower cost.

🏊 Swimming Pools

SCIP used for pool walls — watertight, no termite risk, faster than block, can be sculpted and stamped in any finish. Structurally superior to traditional forms.

🧱 Retaining & Sound Walls

Excellent for highway sound walls, retaining walls, and perimeter security. Wind-rated, seismically rated, fast installation, finishable in any texture.

🏗️ Industrial & Federal

Bullet-resistant (NIJ Level IIIA) — used in federal buildings. Military housing projects. Warehousing. Cold storage (insulated core is a bonus).

Installation: Faster Than You Think

From panel delivery to finished walls — here's the timeline that amazes contractors

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01

Foundation & Footings

⏱ Standard

No special foundation required. Panels anchor to standard concrete footings with rebar dowels. Same as conventional construction.

02

Panel Delivery & Staging

⏱ Day 1–2

Pre-fabricated TIPs panels arrive with openings already cut. Lightweight panels (50% lighter than solid concrete) require standard material handling.

03

Panel Erection

⏱ Days 2–5

Panels positioned and secured with connection mesh and rebar at corners and openings. Small crew can erect a complete residential wall system in days, not weeks.

04

MEP Rough-In

⏱ Before Shotcrete

Electrical conduits, plumbing, and HVAC rough-ins are completed inside the panel. Use heat gun to carve EPS channels — easy, clean, fast. No special tools required.

05

Shotcrete Application

⏱ 1–2 Days

High-pressure shotcrete (3,500 psi min.) applied to both sides over wire mesh. Skilled shotcrete crew achieves ¾" minimum over mesh on each face.

06

Finishing

⏱ Standard

Interior: drywall, plaster, or trowel finish directly over shotcrete. Exterior: stucco, EIFS, stone veneer, brick, or stamped concrete patterns — limitless options.

Frequently Asked Questions

The questions everyone has but is afraid to ask — answered honestly

Q: Is SCIP code-approved in Colorado?

Yes. SCIP panels are ICC-ES evaluated and can be permitted under IBC and IRC with engineering design. TIPs panels have been used across all 50 states. Always get local AHJ review — as with any alternative system.

Q: Can I get a conventional mortgage on a SCIP home?

Yes. SCIP homes meet VA, FHA, and HUD thermal requirements and can be financed conventionally. Concrete/steel construction is generally viewed favorably by lenders as more durable.

Q: What do contractors think of SCIP?

Initial learning curve exists. However, crews consistently report faster wall completion once trained. Coons Construction provides crew training — you don't have to figure it out alone.

Q: Is it really cheaper? Be honest.

Material cost is comparable to ICF, slightly above wood frame. BUT: when you factor in lower energy costs, reduced HVAC sizing, less rebar, less concrete, and faster build time — total project cost is very competitive.

Q: What about moisture and mold?

EPS foam is non-organic — mold cannot grow on it. The concrete shell is continuous. Properly installed, SCIP walls are highly moisture-resistant. This is a significant advantage over wood frame.

Q: How long does a SCIP building last?

100+ years. The steel and concrete composite is not subject to wood rot, termite damage, or frame fatigue. Several SCIP buildings in Europe are 50+ years old and performing excellently.

SCIP vs. Traditional Construction

Let's put the myths to rest with a side-by-side reality check

⚠️ *Fun fact: The Three Little Pigs built straw, stick, and brick. The wolf blew down two. SCIP = brick. But faster, lighter, and with better insulation.*

Category	Traditional Wood Frame	SCIP with TIPs	Winner
Structural Strength	Relies on framing + OSB + plywood	Monolithic steel-concrete composite	SCIP ✓
Thermal Performance	R-13 to R-21 w/ thermal bridging	R-16+ continuous, no bridging	SCIP ✓
Fire Resistance	1 hour max (with drywall)	1–4 hours (IBC 722 tested)	SCIP ✓
Wind Resistance	Engineered for ~130–150 mph	Tested to 200–283 mph	SCIP ✓
Seismic Performance	Moderate (depends on details)	High (.4g / ~7.5 Richter)	SCIP ✓
Termite/Pest Risk	High — wood is food	Zero — no organic material in walls	SCIP ✓
Moisture/Mold Risk	High — organic framing	Minimal — EPS + concrete	SCIP ✓
Build Speed	Faster for simple designs	Comparable or faster (pre-fab)	Tie / SCIP
Upfront Cost	Lower material cost	Slightly higher (replaced by savings)	Wood (short-term)
Lifetime Cost	Higher energy + maintenance	Lower energy + 100-year lifespan	SCIP ✓
Design Flexibility	Excellent	Excellent (stamp/carve/finish anything)	Tie

Why Southern Colorado? Why Now?

The market conditions are perfect for SCIP construction in our region

🌡️ Extreme Colorado Climate

Temperature swings of 80°F+ in a single day. High-altitude UV exposure. SCIP's continuous insulation is purpose-built for these conditions.

🌬️ High Wind Corridor

Southern Colorado regularly sees gusts of 80–100+ mph. SCIP's 200–283 mph wind rating makes it the obvious choice for Pueblo, CS, and mountain communities.

🔥 Wildfire Zone

Concrete/steel exterior = non-combustible. Extremely relevant for areas adjacent to open terrain. Insurance companies take note.

📈 Construction Cost Pressure

Colorado construction costs: $200–$500/sq ft (2025). SCIP's labor and material efficiencies are increasingly competitive as wood frame costs rise.

🏠 Workforce Housing

Colorado faces a significant housing shortage. SCIP's faster build timeline and ADU versatility are direct solutions to the market demand.

🌿 Energy Code Compliance

Colorado's increasingly strict energy codes align well with SCIP's R-16+ mass-wall performance and airtight building envelope.

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What Happens Next?

You've seen the data — here's how to put it to work on your next project

Coons Construction is Southern Colorado's dedicated SCIP builder — and every project we take on is powered by TIPs panels from Morgan, Utah.

01
Schedule a Free Project Consultation

Bring your site plan, lot address, or just an idea. David will walk through SCIP feasibility, rough cost ranges, and TIPs panel requirements for your specific project type.

02
Request a TIPs Panel Specification Quote

Once your design is roughed out, we'll submit plans to TIPs in Morgan, Utah for a material quote — covering panel count, pre-fab openings, and delivery to your Colorado job site.

03
Review Your Energy & Cost Model

We'll model your projected energy savings, HVAC sizing reduction, rebar/concrete savings, and estimated payback period — so you go to your lender with real numbers, not hopes.

04
Connect with the SCIP Community

Stay in the loop: follow Coons Construction for SCIP education updates, project announcements, and Colorado-specific resources as we build our presence in the region.

05
Refer a Builder or Investor

Know a contractor, developer, or architect who should understand SCIP? Send them our way. Education is how we grow this market — and a better-built Colorado benefits everyone.

06
Download This Presentation

This deck — with all sources, references, and cost data — is yours to keep. Share it with your team, your architect, or your lender. The more people understand SCIP, the better.

Why Work with Coons Construction?

We don't just sell panels — we deliver complete SCIP builds from design to move-in

Exclusive TIPs Partnership

All SCIP panels for our projects are supplied by TIPs (Total Integrated Panel Systems) from Morgan, Utah — the most advanced SCIP manufacturer in North America.

Full-Service SCIP Builder

We handle design consultation, permitting support, panel ordering, installation oversight, and shotcrete coordination — one point of contact, zero confusion.

Southern Colorado Experts

Based in Colorado, built for Colorado. We understand local soils, climate, code requirements, and the specific demands of high-altitude and high-wind construction.

Investor-Focused Solutions

We speak your language. SCIP ROI, energy modeling, rental income projections, and portfolio-scale efficiency — we're your construction partner, not just your contractor.

Training & Education

We train our crews and yours. Industry education is how we grow the SCIP market in Colorado — a better-informed investor community builds better projects for everyone.

Proven System, Local Expertise

SCIP has 60+ years of global track record. TIPs has the strongest US system. Coons Construction brings it home to Southern Colorado with boots-on-ground experience.

COONS CONSTRUCTION

Let's Build Something That Lasts.

SCIP · TIPs Panels · Coons Construction Inc

📞 Schedule a Free Consultation

Talk to David about your next investment project

📋 Request a TIPs Panel Quote

Get real numbers for your specific project

📫 Stay Connected

Follow Coons Construction for SCIP updates and project news

📥 Access Additional Resources

This presentation, TIPS literature, ICC reports, FAQs

Building Stronger. Building Smarter. Building for Generations

LET'S BUILD SOMETHING THAT LASTS.

Thank you for your time and interest.
I look forward to answering your questions and exploring how SCIP construction can benefit your next project.

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LOWER COSTS TO OWN

BETTER VALUE OVER TIME